The Geometric Worldline Field Model (GWFM)

$4pt]AGeometricInterpretationofElectronStability$, Superposition, and Field Emergence

Jason Dutton
Independent Researcher

August 2026

Abstract

The Geometric Worldline Field Model (GWFM) proposes that the electron field of quantum field theory arises from a single continuous worldline structure embedded in spacetime. This worldline spans all allowed trajectories, producing a dense presence-density that appears as a continuous field. Forward-time segments correspond to electrons; backward-time segments correspond to positrons. Quantum interference results from geometric overlap of worldline segments, while measurement corresponds to boundary-condition selection of specific segments. GWFM preserves all predictions of quantum mechanics and QFT while providing a geometric interpretation of superposition, identity, and field emergence.

1 Introduction

Quantum mechanics and QFT accurately predict experimental outcomes but leave open the ontological status of particles, fields, and superposition. GWFM begins with a geometric premise: a continuous worldline structure spans spacetime, and the electron field is the projection of this structure. Superposition, interference, and measurement arise from the geometry of this worldline rather than from separate interpretive rules.

2 Core Principles of GWFM

P1 — Continuous Worldline Entity

The electron and positron field arise from a single continuous worldline structure.

P2 — Time Orientation

Forward-time segments manifest as electrons; backward-time segments manifest as positrons.

P3 — Dense Trajectory Ensemble

The worldline spans all allowed paths, forming a dense geometric structure.

P4 — Field Emergence

The electron field is the presence-density of the worldline ensemble.

P5 — Interference as Overlap

Interference patterns arise from geometric overlap of worldline segments.

P6 — Measurement as Boundary Selection

Measurement restricts the allowed worldline segments.

P7 — Identity as Global Geometry

Electron identity is global and geometric, not local or discrete.

3 Mathematical Framework

3.1 Worldline Foundation

Let spacetime be a manifold M with metric $g_{\mu\nu}$. The entity's worldline structure is represented by an ensemble of paths $y(\tau)$ embedded in M. A classical relativistic worldline action is

$$S_{\mathrm{WL}} = -m \int d\tau \sqrt{-g_{\mu\nu}(y(\tau))\, \dot{y}^{\mu}(\tau) \dot{y}^{\nu}(\tau)}. \tag{1}$$

3.2 Spin Structure

Spin degrees of freedom are encoded using Grassmann variables $\psi^{\alpha}(\tau)$:

$$S_{\mathrm{spin}} = \int d\tau\, \psi_{\alpha}(\tau) \dot{\psi}^{\alpha}(\tau). \tag{2}$$

3.3 Electromagnetic Gauge Coupling

Coupling to an electromagnetic potential $A_{\mu}(x)$ is

$$S_{\mathrm{EM}} = q \int d\tau\, \dot{y}^{\mu}(\tau) A_{\mu}(y(\tau)). \tag{3}$$

3.4 Total Worldline Action

$$S_{\text{entity}}[y, \psi] = S_{\text{WL}}[y] + S_{\text{spin}}[\psi] + S_{\text{EM}}[y]. \tag{4}$$

3.5 Presence Density

The presence density $\rho(x)$ is defined as

$$\rho(x) = \int \mathcal{D}y\, \mathcal{D}\psi\; e^{iS_{\text{entity}}[y,\psi]}\, \delta^{(4)}(x - y(\tau)). \tag{5}$$

3.6 Emergent Field

An emergent field $\phi(x)$ can be defined as

$$\phi(x) = \int d\tau\, \delta^{(4)}(x - y(\tau)). \tag{6}$$

with appropriate spinor structure added via ψ^α. The standard Dirac Lagrangian

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi \tag{7}$$

is recovered as the effective field theory of the entity's worldline.

3.7 Curved Spacetime Compatibility

In curved spacetime, the covariant derivative is

$$D_\mu = \partial_\mu + \Omega_\mu + iqA_\mu, \tag{8}$$

where Ω_μ is the spin connection. GWFM is compatible with this structure.

4 Interpretation

4.1 One Entity, Many Appearances

The electron field is the visible projection of one entity's full path structure. Localized particle detections are local manifestations of the worldline.

4.2 Interference as Path Overlap

Interference patterns arise where different segments of the worldline overlap in spacetime.

4.3 Measurement as Path Selection

Measurement imposes boundary conditions that restrict the allowed paths.

4.4 Electrons and Positrons

Forward-time segments appear as electrons; backward-time segments appear as positrons.

4.5 Spin, Charge, and Fields

Spin arises from Grassmann variables and geometric rotation. Charge is associated with topological invariants of the worldline structure.

5 Superposition and Double-Slit

5.1 Superposition as Geometry

Superposition corresponds to the worldline spanning multiple allowed trajectories.

5.2 Double-Slit Experiment

The worldline passes through both slits; the interference pattern is the projection of the worldline ensemble.

6 Geometric Foundations

6.1 Entity as Geometric Object

The worldline is a geometric set whose projection yields the electron field.

6.2 Superposition as Geometric Union

$$\mathcal{T} = \mathcal{T}_A \cup \mathcal{T}_B. \tag{9}$$

6.3 Interference as Intersection

$$I(x) = \text{measure}(\mathcal{T}_A \cap \mathcal{T}_B). \tag{10}$$

6.4 Time Reversal as Orientation

Backward-time segments correspond to

$$\frac{dt}{d\tau} < 0. \tag{11}$$

6.5 Spin as Geometric Rotation

Spin is rotation in an internal fiber bundle attached to the worldline.

6.6 Charge as Topological Invariant

Charge may be associated with winding number or Chern classes.

6.7 Measurement as Boundary Geometry

Measurement is the intersection of the worldline with detector boundary geometry.

6.8 Entanglement as Unified Geometry

Entangled systems correspond to a single geometric object spanning multiple regions.

7 Appendix A: Mathematical Derivations

Worldline action, path integral measure, presence density, spinor construction, and Dirac field emergence.

8 Appendix B: Extended Structures

Speculative consciousness content removed. All physical extensions preserved.

9 Appendix C: Superposition — Deep Interpretation

Superposition as distributed manifestation of one worldline across multiple trajectories.